# MATH MA'AM ACADEMY

*Where Every Subject Clicks.*

# STD 5 MATHEMATICS – CHAPTER 8

# DATA HANDLING – NOTES

**Learning objectives:** Understand data, organise information, read pictographs, interpret bar graphs and understand line graphs.

## 8.1 DATA

**Data** is a collection of facts or information gathered for a particular purpose. Data may be collected by asking questions, observing, counting or measuring.

**Example:** The number of students choosing different sports, the daily temperature of a city, or the number of books borrowed from a library are all examples of data.

### Organising data

Raw data is easier to understand when it is arranged in a systematic form. A **frequency table** lists each category or value and the number of times it occurs. The number of occurrences is called its **frequency**.

| Favourite pet | Number of students (frequency) |
|---|---|
| Cat | 8 |
| Dog | 6 |
| Fish | 4 |
| Bird | 2 |

### Tally marks

Tally marks are quick strokes used for counting. The fifth stroke crosses the previous four strokes to make a group of five. For example, 13 is recorded as two groups of five and three extra strokes: **5 + 5 + 3 = 13**.

## 8.2 REPRESENTATION OF DATA

Data can be represented using pictures, bars or lines. A suitable representation makes information easier to compare and understand.

### A. Pictograph

A **pictograph** represents data using pictures or symbols. Each picture stands for a fixed number of items. The **key** tells us the value of one picture.

**Example:** If one apple symbol represents 5 apples and Monday has 6 symbols, then apples sold on Monday = 6 × 5 = **30 apples**.

### How to read a pictograph

1. Read the title to know what the data represents.
2. Read the key carefully before counting symbols.
3. Count the full symbols and multiply by the key value.
4. If a half-symbol is used, take half the key value.
5. Add or subtract category totals when required.

**Important: Never assume that one picture always represents one item. Always check the key.**

### B. Bar graph

A **bar graph** represents data using rectangular bars. Bars may be vertical or horizontal. Each bar has equal width, and the height or length of a bar represents the value for that category. Bars are generally separated by equal gaps.

**Example data for a bar graph:**

| Fruit | Number sold |
|---|---|
| Apples | 12 |
| Bananas | 8 |
| Oranges | 15 |
| Grapes | 5 |

**Reading the table:** Oranges have the highest count (15), grapes have the lowest (5), and total fruit sold = 12 + 8 + 15 + 5 = **40**.

### Steps to read a bar graph

1. Read the title and category labels.
2. Check the scale on the vertical or horizontal axis.
3. Match each bar to its category.
4. Read the value from the scale.
5. Compare bars, find totals or calculate differences as asked.

**Remember: The scale may increase by 1, 2, 5, 10 or another equal interval. Do not assume every small division represents 1.**

### C. Line graph

A **line graph** represents data by plotting points and joining them with straight line segments. It is especially useful for showing how data changes over time, such as daily temperature, rainfall or sales.

### How to read a line graph

1. Read the title and labels on both axes.
2. Check the scale on the vertical axis.
3. Locate the required day, time or category on the horizontal axis.
4. Read the value at the plotted point.
5. Compare points and observe whether the line rises, falls or stays level.

### Meaning of line segments

- **Rising line:** the value is increasing.
- **Falling line:** the value is decreasing.
- **Horizontal line:** the value remains unchanged.

## Choosing the correct representation

| Representation | Best used for |
|---|---|
| Pictograph | Showing quantities with pictures or symbols. |
| Bar graph | Comparing quantities in different categories. |
| Line graph | Showing changes or trends over time. |
| Frequency table | Listing categories and their counts. |

## Quick revision formulas and checks

- Total = Add all category values.
- Difference = Larger value – Smaller value.
- Pictograph total = Number of symbols × Value of one symbol.
- Number of symbols = Total quantity ÷ Value of one symbol.
- Half-symbol value = Value of one full symbol ÷ 2.

## Common mistakes to avoid

- Forgetting to read the pictograph key.
- Reading the wrong scale on a bar or line graph.
- Comparing bar heights without checking axis intervals.
- Confusing the highest value with the total value.
- Forgetting units such as students, books, cm or °C in the final answer.

**Study tip: Practise reading the title, key, labels and scale first. Then answer questions about highest, lowest, total, difference and change.**

**ALL THE BEST**